

Chapter 2 形式邏輯 Formal Logic

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2.1 命題邏輯 Propositional Logic

Propositional Variables 命題變項	Atomic Propositions 原子命題	Compound Propositions 複合命題
簡單定義: 邏輯學家的「代號空箱子」, 用來隨便裝任何一句話。 Simple Definition: The "empty code box" of logicians, used to hold any arbitrary sentence. 形式代號: 通常寫成 p 或者 q。 Formal Symbol: Usually written as p or q.	簡單定義: 不能再切一刀的「最底層事實單元」, 裡面不包含任何「並且」、「如果」、「並非」的黏合劑。 Simple Definition: The "bottom-level factual unit" that cannot be cut any further, containing zero logical glues like "and," "if," or "not." 「這盒牛奶是過期的。」(這是一句最簡單的陳述, 拆不開了, 這就是原子命題)。 "This milk carton is expired." (This is the simplest statement, it cannot be broken apart; this is an atomic proposition). 以為「這盒牛奶不是過期的」也是原子命題。不對! 因為它裡面藏了一個「不是(並非)」, 這已經動了刀子、加了化學劑了。 Thinking "This milk carton is not expired" is also an atomic proposition. Wrong! Because it hides a "not," which means an operation has already been applied.	簡單定義: 把好幾顆原子積木, 用「邏輯黏合劑(聯結詞)」拼裝出來的「變形金剛大命題」。 Simple Definition: A "transformer proposition" assembled by gluing several atomic blocks together using "logical glues (connectives)." 原子命題 p = 這盒牛奶是過期的。 Atomic proposition p = This milk carton is expired. 原子命題 q = 這家店會被罰款。 Atomic proposition q = This shop will be fined. 我們用黏合劑把它們連起來: Let's glue them together: 「如果這盒牛奶是過期的, 那麼這家店會被罰款。」(這句話就變成了複合命題, 公式寫作 $p \rightarrow q$)。 "If this milk carton is expired, then this shop will be fined." (This sentence becomes a compound proposition, formulated as $p \rightarrow q$).

Logical Connectives 邏輯連結詞			
	Standard Symbol 標準符號	Simple Definition 簡單定義	Example 例子
Negation 否定	(讀作 "Not" / 「並非」)	邏輯唱反調，直接把本來的對錯(真假)完全顛倒。 Logical opposition; it completely reverses the original truth value.	Let p = 這批牛奶含有防腐劑。 令 p = 這批牛奶含有防腐劑。 $\neg p$ = 並非這批牛奶含有防腐劑 (這批牛奶不含防腐劑)。 $\neg p$ = It is not the case that this batch contains preservatives.
Conjunction 聯言/合取	(讀作 "And" / 「並且」)	極度嚴格的「同時成立」條款，兩邊必須都是真的，整句話才算數。 An extremely strict "simultaneous" clause; both sides must be true for the whole statement to hold.	「這盒牛奶是當天現擠的 (p) 並且這盒牛奶通過了安全檢測 (q)。」 "This milk is freshly squeezed today (p) and this milk passed the safety inspection (q)." (公式化寫作: $p \wedge q$ 。只要其中有一條是假的，比如雖然是現擠的但沒過檢，這整句承諾就當場破產。) (Formulated as: $p \wedge q$. If even one part is false—like being fresh but failing inspection—the whole promise goes bankrupt.)
Disjunction 選言/析取	(讀作 "Or" / 「或者」)	寬容度極高的「二選一或全選」條款，只要有一邊是真的，整句話就成立。 A highly tolerant "pick one or take both" clause; as long as at least one side is true, the whole statement holds.	「我們公司將賠償你一千元 (p) 或者我們公司將補償你十箱新牛奶 (q)。」 "Our company will compensate you \$1000 (p) or our company will compensate you with 10 cartons of new milk (q)." (公式化寫作: $p \vee q$ 。如果老闆兩個都給了，邏輯上這句話依然是真的，你不能告他撒謊。) (Formulated as: $p \vee q$. If the boss gives you both, logically this statement remains true, and you cannot sue him for lying.)
Material Implication 實質蘊涵/條件	(讀作 "If... then..." / 「如果...那麼...」)	最容易被黑心商人利用的「單向承諾」條款。只有當前半句承諾發生了、後半句卻沒做到時，才算違約。 A "one-way promise" clause most easily exploited by shady merchants. A breach occurs only when the first half happens but the second half fails.	「如果你喝了這款牛奶拉肚子 (p)，那麼我們公司會全額退款 (q)。」 "If you get diarrhea from drinking this milk (p), then our company will give you a full refund (q)." (公式化寫作: $p \rightarrow q$ 。大坑來了：如果你今天根本沒拉肚子 (p 為假)，但公司心情好依然退款給你，或者不退款給你，在邏輯上這份合同都沒有被違反！) (Formulated as: $p \rightarrow q$. Here is the massive trap: if you don't get diarrhea today (p is false), whether the company gives you a refund or not, the contract is logically NOT breached!)
Biconditional 雙條件/等值	(讀作 "If and only if" / 「當且僅當」)	終極的「生死綁定」條款。兩邊必須同生共死(同真或同假)，整句話才算成立。 The ultimate "ride-or-die binding" clause. Both sides must stand or fall together (both true or both false) for the statement to hold.	「我們公司收購你的牧場 (p)，當且僅當你承諾不再提供奶源給對手 (q)。」 "Our company buys your farm (p) if and only if you promise to stop supplying milk to our rivals (q)." (公式化寫作: $p \leftrightarrow q$ 。如果你沒給對手提供奶源，我們就必須收購；如果你提供了，我們就絕對不收購。兩邊的命運被牢牢焊死了。) (Formulated as: $p \leftrightarrow q$. If you don't supply rivals, we must buy; if you do supply them, we absolutely won't. The fates of both sides are welded solid.)

2.2 Truth Tables 真值表

Truth Value Assignments & Binary Logic Values 真值指派 與 二值邏輯值	商戰例子 Business Example
簡單定義：邏輯世界裡只有兩種死狀態：對（真）、或者錯（假）。而「指派」，就是我們在模擬市場情況時，假設某句話是對的還是錯的。 Simple Definition: The logical world has only two strict states: True or False. "Assignment" is when we simulate market scenarios by assuming a statement is either true or false.	We have two text codes: 我們手裡有兩個文字代號： • Code A = The milk is expired. (代號甲 = 牛奶過期了。) • Code B = The clerk ran away. (代號乙 = 店員跑路了。) Now we do a "truth value assignment" by listing all possibilities like an accountant: 現在我們進行「真值指派」，就像會計做賬一樣，把所有可能性列出來： • Scenario 1: Code A is true, Code B is true. (情況一：代號甲是真的，代號乙也是真的。) • Scenario 2: Code A is true, Code B is false. (情況二：代號甲是真的，代號乙是假的。) • Scenario 3: Code A is false, Code B is true. (情況三：代號甲是假的，代號乙是真的。) • Scenario 4: Code A is false, Code B is false. (情況四：代號甲是假的，代號乙是假的。)

Classification of Formulas 公式分類		
1. Tautology (Absolute Defense) 永真式(絕對防禦)	2. Contradiction (Instant Suicide) 矛盾式(當場自殺)	3. Contingency (Real World Truth) 偶真式(人間真實)
The statement is logically always true, no matter what happens in the market. 無論市場發生什麼鬼情況，這句話在邏輯上永遠都是對的。 The owner says: "Either this milk is expired, or it is not expired." 老闆說：「這盒牛奶要麼是過期的，要麼不是過期的。」 This sentence is just a cliché; no matter the true state of the milk, the statement is always true. 這句話就是句廢話，無論牛奶到底過期沒，整句話永遠是對的。	The statement is always false under any circumstances; it's slapping one's own face. 無論在什麼情況下，這句話聽起來都一定是錯的，屬於自己打自己臉。 The owner says: "This milk is fresh and not fresh at the same time." 老闆說：「這盒牛奶同時既是新鮮的，又並非是新鮮的。」 He strangled his own words; no matter the truth, this sentence is always false. 他自己把自己的話給掐死了，不管牛奶真相如何，這句話永遠是假的。	Sometimes true, sometimes false, depending on the actual reality. 99% of business contract clauses are contingencies. 有時候是對的，有時候是假的，得看現實情況。商業合約裡 99% 的話都是偶真式。 The owner says: "If the milk is expired, then the clerk will run away." 老闆說：「如果牛奶過期，那麼店員就會跑路。」 The truth value is uncertain. If it is expired but the clerk didn't run, the owner lied; under other scenarios, he is right. 這句話的對錯是不一定的。如果牛奶真過期了而店員沒跑路，那老闆就在撒謊；在其他情況下，這句話算他對。

Logical Equivalence Verification

邏輯等價驗證

Checking if two contract clauses with different wordings produce identical true/false columns in a truth table. If yes, they are "equivalent", meaning you can swap them to shatter your opponent's word games.

檢查兩份不同字面的合同條款，其真值表的對錯結果是不是完全複製貼上。如果是，這兩句話就叫「等價」，你可以隨時拿來替換，拆穿對手的文字遊戲。

De Morgan's Laws 德摩根定律

The opponent's blurry contract: "It is not the case that (the milk is freshly squeezed AND the price is cheap)."

對手的模糊合約：「並非（牛奶是現擠的 並且 價格是便宜的）」

Your truth table equivalent conversion: "The milk is not freshly squeezed OR the price is not cheap."

你用真值表等價轉換後：「牛奶不是現擠的 或者 價格不是便宜的」

De Morgan's Laws tell you that when distributing "not" into brackets, the "AND" inside must automatically flip into "OR". This is the fastest mirror to expose slippery players when they write super complex clauses.

德摩根定律告訴你，把外面的「並非」拆進括號裡時，括號裡面的「並且」必須要自動反轉變成「或者」。這是對付老狐狸把合約寫得極其複雜時，最快現形的照妖鏡。

Distributive Laws 分配律

The opponent's forced clause: "The milk is freshly squeezed, AND (it passed inspection OR it got a certificate)."

對手的霸王條款：「牛奶是現擠的，並且（通過了安檢 或者 拿到了證書）」

Your truth table equivalent conversion: "(The milk is freshly squeezed AND passed inspection) OR (The milk is freshly squeezed AND got a certificate)."

你用真值表等價轉換後：「（牛奶是現擠的 並且 通過了安檢）或者（牛奶是現擠的 並且 拿到了證書）」

This acts just like the distributive law of multiplication in algebra. You can distribute the block "freshly squeezed and" directly to everyone inside the brackets. Whether packing or unpacking, the scores on the truth table are identical.

這就像數學裡的乘法分配律。你可以把「並且是現擠的」這團膠水，直接分發給括號裡面的每一個人。不管是打包還是拆包，這兩句話在真值表上的得分完全一模一樣。

2.3 Rules of Inference 推理規則

Modus Ponens (The Straight Punch) 肯定前件 (正拳直擊)	Modus Tollens (The Reverse Counter) 否定後件 (絕地反擊)	Hypothetical Syllogism (The Chain Reaction) 假言三段論 (多米諾骨牌)	Disjunctive Syllogism (The Elimination Trap) 選言三段論 (關門捉賊)	Constructive Dilemma (The Double Fork Trap) 構成性兩難 (雙重夾擊)
Simple Definition: If the "If-clause" happens, the "Then-clause" automatically triggers. 人話定義：如果前半句承諾的條件觸發了，後半句的結果就必須給我交出來。	Simple Definition: If the expected result did not happen, it means the trigger condition never existed. 人話定義：如果後半句的結果根本沒發生，那就倒推證明前半句的條件完全是瞎編的。	Simple Definition: Linking promises together. If A leads to B, and B leads to C, then A directly leads to C. 人話定義：連鎖反應。如果事情甲會引發事情乙，而事情乙又必定會引發事情丙，那事情甲就直接鎖定了事情丙。	Simple Definition: An "either-or" scenario where you eliminate one option, forcing the other to be the absolute truth. 人話定義：面臨二選一的局面，你只要拿證據拍死其中一個選項，另一個選項就自動變成唯一真相。	Simple Definition: Forcing your opponent into two paths, where both paths lead to an execution. 人話定義：給對手設下兩條路，不管他挑左邊還是挑右邊，終點都安排了一把斷頭台等著他。
The contract states: "If this milk expires, then the shop pays a fine." 合約規定：「如果牛奶過期，那麼店家就要付罰款。」 Now, you prove: "This milk is expired!" 現在，你拿出證據證明：「這盒牛奶確實過期了！」 Unstoppable Conclusion: "Therefore, the shop must pay the fine!" 無可辯駁的結論：「所以，店家必須立刻付罰款！」	The merchant claims: "If our milk contains toxins, your stomach will hurt instantly." 商家吹牛說：「如果我們的牛奶有毒，你喝了肚子會立刻痛。」 You counter: "Look, my stomach does not hurt at all after drinking your competitor's milk." 你反擊說：「看好了，我喝了你對手的牛奶，肚子完全不痛。」 Unstoppable Conclusion: "Therefore, your milk is not the one I drank! (Or: Your premise about the symptom is totally fake.)" 無可辯駁的結論：「所以，你口中那個『只要有毒就會痛』的因果關係被我打破了！」	Rule 1: "If the farm cows get sick, the milk supply drops." 規則一：「如果牧場的母牛生病，牛奶供應量就會下跌。」 Rule 2: "If the milk supply drops, our store stock price crashes." 規則二：「如果牛奶供應量下跌，我們店的股價就會暴跌。」 Unstochastic Conclusion: "Therefore, if the farm cows get sick, our store stock price crashes." 無可辯駁的結論：「所以，只要牧場母牛一聲咳嗽，我們店的股價就注定暴跌。」	The manager says: "Either the manager stole the milk recipe, or the clean up guy did it." 經理說：「要麼是副經理偷了牛奶配方，要麼是清潔工幹的。」 The police prove: "The clean up guy was on vacation in Paris and has a perfect alibi!" 警方證實：「清潔工當時正在巴黎度假，有完美的不在場證明！」 Unstoppable Conclusion: "Therefore, the manager did it!" 無可辯駁的結論：「所以，不用查了，配方就是副經理偷的！」	You set up the premises: 你設下致命局勢： Path 1: "If you admit you sold fake milk, you must pay a fine." 路徑一：「如果你承認賣假奶，你必須賠錢。」 Path 2: "If you lie about it, we will call the cops and you go to jail." 路徑二：「如果你不承認而撒謊，我們會報警把你抓去坐牢。」 The trap: "You must either admit it now or speak a lie under our cross-examination." 夾擊前提：「你現在要麼乖乖承認，要麼在我們的交叉盤問下繼續撒謊。」 Unstoppable Conclusion: "Therefore, you will either pay a fine or go to jail!" 無可辯駁的結論：「所以，你今天要麼破財，要麼坐牢，你挑一個吧！」

Rules of Replacement 替換規則			
Double Negation (The Clarifier) 雙重否定 (負負得正)	Commutativity (The Order Flipper) 交換律 (左右互換)	Associativity (The Group Shifter) 結合律 (括號大挪移)	Transposition (The Reverse Logic Trap) 位換律 (逆否命題轉換)
Simple Definition: Saying "not-not X" is exactly the same as saying "X". It cuts through verbal gymnastics. 人話定義：「並非不成立」就是「成立」。用來直接剪掉對手故意繞圈子的雙重廢話。	Simple Definition: For "AND" or "OR" statements, the left and right sides can be swapped freely. The order does not change the deal. 人話定義：只要全句是用「並且」或「或者」連接的，左邊和右邊的順序隨便你換，合約效力完全複製貼上。	Simple Definition: When a long statement uses only "AND" (or only "OR"), you can move the brackets to group items however you like. 人話定義：當一句長話裡面全部都是「並且」(或全部都是「或者」)時，括號要先包庇誰、先綁定哪兩件事，都完全隨便你。	Simple Definition: "If A then B" is 100% equivalent to "If NOT-B then NOT-A". This is the most powerful weapon to catch contract dodgers. 人話定義：「如果甲，那麼乙」可以完美變形成「如果不是乙，那麼就絕對沒有甲」。這是抓合約漏洞最強大的變形術。
The shady merchant says: "It is not true that our milk is not inspected." 黑心商家狡辯：「說我們的牛奶沒有經過檢驗，這並非事實。」 You instantly replace it with: "So, you are claiming your milk IS inspected." 你當場把它重寫為：「所以，你就是堅稱你的牛奶『有』經過檢驗。」	The clause says: "You get a refund (A) AND we take back the spoiled milk (B)." 條款寫著：「你獲得退款(甲)，並且 我們收回壞掉的牛奶(乙)。」 You can replace it with: "We take back the spoiled milk (B) AND you get a refund (A)." 你完全可以替換為：「我們收回壞掉的牛奶(乙)，並且 你獲得退款(甲)。」	The contract states: "Milk is fresh, AND (it's safe AND it's cheap)." 合同規定：「牛奶是新鮮的，並且(它是安全的 並且 它是便宜的)。」 You can move the brackets and rewrite it as: "(Milk is fresh AND it's safe), AND it's cheap." 你完全可以挪動括號，重寫為：「(牛奶是新鮮的 並且 它是安全的)，並且 它是便宜的。」	The original promise: "If our company is a top-tier brand (A), then our milk safety is perfect (B)." 對方原本的吹牛：「如果我們公司是頂級品牌(甲)，那麼我們的牛奶安全性就是完美的(乙)。」 You use Transposition to rewrite their own rule: "If your milk safety is NOT perfect (not-B), then your company is NOT a top-tier brand (not-A)." 你直接用位換律重寫他們的家規：「如果你們的牛奶安全性不完美(不是乙)，那麼你們公司就絕對不是頂級品牌(不是甲)。」

2.4 Predicate Logic

謂詞邏輯

First-Order Logic 一階邏輯	Individuals, Constants, and Variables 個體、常項 與 變項	Predicates and Relations 謂詞 與 關係
	Simple Definition: Individuals are your data entries. Constants are real, specific names. Variables are just the blank search boxes. 人話定義：個體就是數據庫裡的資料。常項是具體的真名（如：那頭叫花花的母牛）；變項只是搜尋欄裡的空白方塊（如：某個叫「未知對象」的東西）。 Constant (Real name): "Farm-A". It targets one specific factory. 常項（真名）：「A號牧場」。指的就是那家特定工廠，不能指別人。 Variable (Blank box): "Factory-X". It can mean any factory in the country until we fill it in. 變項（空白方塊）：「X號工廠」。它可以指全國任何一家工廠，看你待會把誰的名字填進去。	Simple Definition: A predicate is the "Status/Tag" attached to a name. A relation is the "Arrow" connecting two names. 人話定義：謂詞就是貼在名字後面的「屬性標籤」（如：是壞掉的）；關係就是連接兩個名字的「互動箭頭」（如：甲欺騙了乙）。 Predicate (Tag): "The milk (X) [is expired]". We are checking the quality of X. 謂詞（標籤）：「牛奶（未知對象）[是過期的]」。我們在描述這個對象的屬性。 Relation (Arrow): "[Company-A] bought milk from [Farm-B]". It connects two specific business entities. 關係（箭頭）：「[A公司] 從 [B牧場] 採購牛奶」。這在描述兩個主體之間的交易動作。
Quantifiers 量詞	Universal Quantifier 全稱量詞	Existential Quantifier 存在量詞
	The Business Meaning: The ultimate statement of corporate overconfidence. It claims 100% perfection with zero exceptions. 商業內涵：企業盲目自信的極致體現。它宣稱的是 100% 的完美，不留任何退路與例外。 The Catchphrase: "All...", "Every...", "Always...", "Never...". 經典口頭禪：「所有.....」、「每一個.....」、「總是.....」、「從不.....」。	The Business Meaning: The sniper rifle of the auditor or prosecutor. It doesn't care about the average; it looks for the smoking gun. 商業內涵：審計師或檢察官手裡的狙擊槍。它不在乎平均表現有多好，它只盯著 那個致命的犯罪鐵證。 The Catchphrase: "There exists...", "At least one...", "Some...", "A certain...". 經典口頭禪：「存在.....」、「至少有一個.....」、「某些.....」、「某個.....」。

The Four Quantification Rules (The Legal Investigation)

四大量化規則(法庭調查實戰)

These four rules tell you how to legally move between a "General Target" and a "Specific Suspect".
這四個規則教你如何在法律上，把「大範圍目標」與「具體嫌疑人」進行不可抵賴的轉換。

1. Universal Instantiation (Locking the Suspect) 全稱實例化(關門抓人)	2. Universal Generalization (Writing the Code) 全稱概括(製定天條)	3. Existential Instantiation (Giving the Mole a Name) 存在實例化(給內鬼代號)	4. Existential Generalization (Securing the Conviction) 存在概括(定罪蓋章)
How it works: If the rule applies to everyone, it applies to you. 人話定義：既然地規規定所有人都逃不掉，那你這個具體的人也別想跑。	How it works: If a randomly inspected batch with no privileges passes, the whole line is cleared. 人話定義：如果我們隨機抽樣一個沒有任何特殊優待的標準品，發現它完全合格，那我們就能對整條生產線下全面合格的結論。	How it works: We know a thief exists; let's call him "Suspect-A" for the investigation. 人話定義：我們明知道團隊裡一定有一個人泄密了，在查出真名之前，我們先把他代號定為「嫌疑人甲」，展開追查。	How it works: If we catch a specific guy doing it, we have proven that the crime exists. 人話定義：只要我們當場抓到某個具體的人在投毒，我們就成功向法官證明了「這家工廠確實存在投毒行為」。
Premise: "ALL executives who sign fake financial logs must go to jail." 大前提：「【所有】在假賬本上簽字的高管，通通都要坐牢。」 Your investigation: "The CEO signed this fake financial log." 法庭調查：「執行長，你在這本假賬上簽了字。」 Conclusion: "Therefore, the CEO must go to jail." 結論：「所以，執行長你準備去坐牢吧。」	You test a randomly selected, normal individual bottle: "This random Bottle-X from the conveyor belt is contaminant-free." 你抽檢了一個毫無特權的普通瓶子：「這條傳送帶上的隨機X號瓶，沒有任何化學殘留。」 Conclusion: "Therefore, ALL milk bottles on this assembly line are contaminant-free." 結論：「所以，這整條生產線產出的【所有】牛奶，通通都沒有化學殘留。」	The ledger audit reveals: "There EXISTS a person who stole ten million from the milk fund." 賬目審計顯示：「賬面上【少了一千萬】，團隊裡絕對存在一個小偷。」 You activate the rule: "Let's call this mysterious thief 'Suspect-A' and track his bank logs." 你啟動規則：「我們把這個神秘小偷代稱為『嫌疑人甲』，調閱這個代號背後的所有轉賬紀錄。」	Evidence secured: "Manager John was caught red-handed pouring toxic bleach into Tank-3." 證據確鑿：「經理約翰被當場抓到往3號儲水池裡倒入有毒漂白水。」 Conclusion: "Therefore, there EXISTS an employee in this factory who is poisoning the product!" 結論：「報告法官，本案正式成立，這家工廠內部【確實存在】員工投毒的犯罪事實！」